OTEL-07: Dynatrace OTLP Integration

> **Series:** OTEL — OpenTelemetry Integration | **Notebook:** 7 of 8 | **Created:** January 2026 | **Last Updated:** 10/29/2026

Complete Setup for OpenTelemetry with Dynatrace

This notebook provides end-to-end configuration for sending OpenTelemetry data to Dynatrace, including authentication, endpoints, and verification.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS or Managed with OTLP enabled

Requirement	Details
Permissions	Token creation access
Knowledge	OTEL-01 through OTEL-06

SaaS 1.337: OpenTelemetry-relevant changes

1. **OTel `service.name` in SDv1 service names.** Per the release notes, *"You can now use the OTel service.name attribute to customize the name of SDv1 services."* Smartscape shows `service.name (detected name)`, and the change *"affects the service name in all Smartscape use cases, and the dt.service.name attribute on spans and metrics."* `dt.service.name` is on every span — OneAgent and OTLP alike — so it is the field to group by:
   `dql fetch spans, from:-1h | filter isNotNull(dt.service.name) | summarize span_count = count(), by:{dt.service.name, dt.smartscape.service}`
2. **OneAgent + OpenTelemetry-injector coexistence matrix** — documented in **K8S-11 § 2a**. Covers double-instrumentation symptoms, init-container ordering, `LD_PRELOAD` / `JAVA_TOOL_OPTIONS` clobber, trace-context conflicts, and a per-namespace decision flow. Read it before planning OTel auto-instrumentation where OneAgent is also deployed.
3. **SDv2 for AWS Lambda (Early Access).** *"It provides a single unified ruleset for both OpenTelemetry and OneAgent instrumentation."* See **CLOUD-04 § Sprint 1.337**.

> **Sources:** SaaS 1.337 release notes (DT docs).

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1. Dynatrace OTLP Endpoints

SaaS Endpoints

Dynatrace supports **OTLP/HTTP only** for native ingest, with binary Protocol Buffers — *"gRPC is not supported. API calls need to use HTTP."* and *"JSON is not supported for Protocol Buffers. Binary format must be used."* Use a Collector to convert gRPC to HTTP.

Signal	HTTP Endpoint
All (base)	`https://{env-id}.live.dynatrace.com/api/v2/otlp`
Traces	`https://{env-id}.live.dynatrace.com/api/v2/otlp/v1/traces`
Metrics	`https://{env-id}.live.dynatrace.com/api/v2/otlp/v1/metrics`
Logs	`https://{env-id}.live.dynatrace.com/api/v2/otlp/v1/logs`

> **Important:** gRPC is **not supported** for direct Dynatrace ingest. If your SDKs use gRPC, route through a Collector with an `otlp` gRPC receiver and `otlphttp` exporter. See Transform OTLP gRPC.

ActiveGate Endpoints

For on-premises or network-restricted environments:

Signal	Endpoint
All	`https://{activegate-host}:9999/e/{env-id}/api/v2/otlp`

Dynatrace Collector Distribution

Dynatrace provides its own **Collector distribution** with verified, production-ready components:

Aspect	Detail
Image	`ghcr.io/dynatrace/dynatrace-otel-collector/dynatrace-otel-collector`
Features	Pre-configured for Dynatrace, upstream-compatible components
Docs	Dynatrace Collector

> **Tip:** Use the Dynatrace Collector distribution for production. Its x86-64 and ARM64 builds *"are supported by the Dynatrace Support team"*. For Core, Contrib and Builder setups Dynatrace gives limited support, and *"This support covers only the components and their versions that are included in the Dynatrace OTel Collector."*

> Contrib is for test setups: *"Dynatrace recommends using the Contrib distribution only for testing purposes."*
>
> If you need a component the Dynatrace distribution does not ship, the same page lists the OpenTelemetry Collector Builder for building your own Collector — *"When you need to fully customize the Collector instance and only run the components required for your use case."* A Builder setup gets the same limited support, so components outside the Dynatrace distribution are not covered.
>
> **Sources:** Dynatrace OTel Collector (DT docs) — *"Dynatrace provides limited support for Core, Contrib, and Builder setups."*

Environment ID

Find your environment ID:
1. Log into Dynatrace
2. Look at the URL: `https://abc12345.live.dynatrace.com`
3. `abc12345` is your environment ID

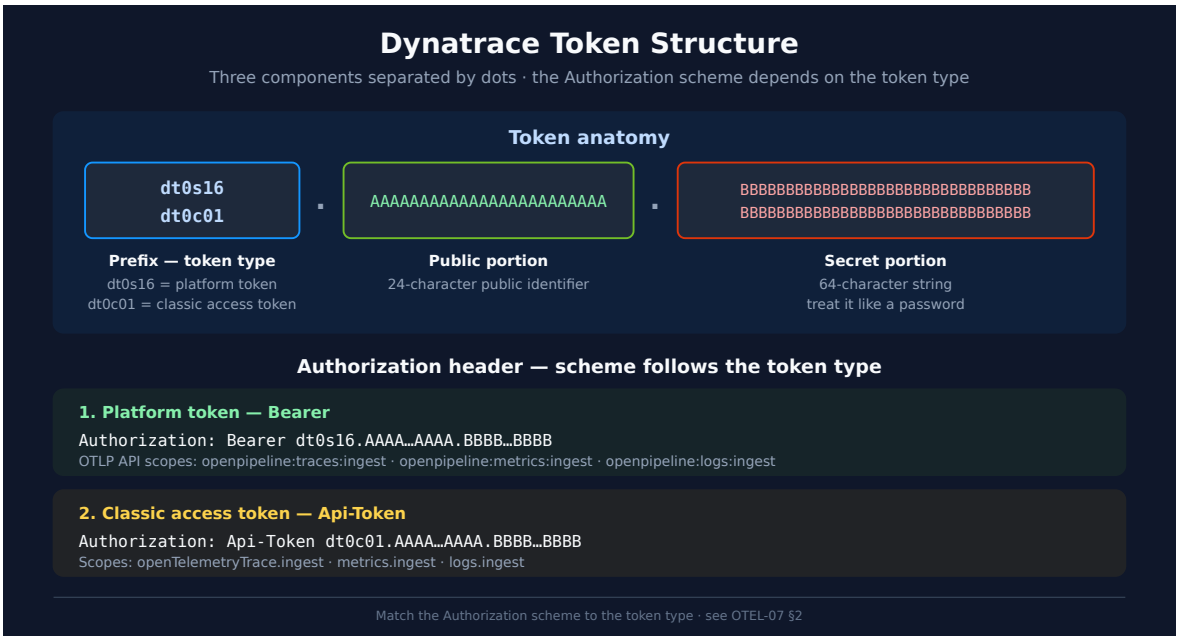
2. Authentication Setup

Create API Token

1. Navigate to **Settings > Access tokens**
2. Click **Generate new token**
3. Name: `otel-ingest-token`
4. Select scopes:

Scope	Signal	Required
`openTelemetryTrace.ingest`	Traces	Yes (for traces)
`metrics.ingest`	Metrics	Yes (for metrics)
`logs.ingest`	Logs	Yes (for logs)

Token Format



Header Format

Use the `Api-Token` prefix in the Authorization header:

```
Authorization: Api-Token &lt;your-full-token&gt;
```

Token Type and Auth Scheme

The `Api-Token` scheme above is for **classic API Tokens** (prefix `dt0c01.*`). **Platform Tokens** (prefix `dt0u01.*` for user tokens, `dt0s16.*` for service-user tokens) use `Bearer ${TOKEN}` instead:

```
# Classic API Token
Authorization: Api-Token dt0c01.XXXXXXXX...

# Platform Token (Gen3-preferred)
Authorization: Bearer dt0u01.XXXXXXXX...
```

Using the wrong scheme returns **401 Unauthorized** even when the token has the correct scopes — a frequent first-deployment failure mode. Platform Tokens are preferred in Gen3 environments because they support fine-grained scopes via IAM policies. For OTLP ingest both token types work, with different scopes: a Platform Token needs

`openpipeline:traces:ingest`, `openpipeline:metrics:ingest` and `openpipeline:logs:ingest`; a classic token needs the scopes in the table above.

> **Sources:** OTLP API (DT docs) — *"To authenticate, use a platform token with `Bearer` in the `Authorization` HTTP header." "The required scopes are `openpipeline:logs:ingest`, `openpipeline:metrics:ingest`, and `openpipeline:traces:ingest`."*; API authentication (DT docs) — `openTelemetryTrace.ingest`: *"Grants permission to ingest OpenTelemetry traces."*

3. Collector Configuration

Image Versioning Best Practice

> **Important:** Always pin your OTel Collector image to a specific version. Using `latest` can cause unexpected behavior during upgrades.

```
# Avoid - Non-deterministic deployments
image: otel/opentelemetry-collector-contrib:latest

# Production - the Dynatrace distribution, pinned (see §1 for what Dynatrace
Support covers)
image: ghcr.io/dynatrace/dynatrace-otel-collector/dynatrace-otel-
collector:v0.48.0

# Test setups - Contrib, pinned (Dynatrace recommends Contrib for testing
only)
image: otel/opentelemetry-collector-contrib:0.151.0
```

Check the OpenTelemetry Collector releases or Dynatrace Collector releases for the current stable version.

Complete Collector Config for Dynatrace

```
# otel-collector-config.yaml
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
      http:
        endpoint: 0.0.0.0:4318

processors:
  batch:
    timeout: 10s
    send_batch_size: 1000

  memory_limiter:
    check_interval: 1s
    limit_mib: 800
    spike_limit_mib: 200

  # Add service.name if missing
  resource:
    attributes:
      - key: service.name
        value: "unknown-service"
        action: insert

exporters:
  otlphttp/dynatrace:
    endpoint: https://${DT_ENV_ID}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}
    retry_on_failure:
      enabled: true
      initial_interval: 5s
      max_interval: 30s
      max_elapsed_time: 300s

  debug:
    verbosity: detailed

extensions:
  health_check:
    endpoint: 0.0.0.0:13133

service:
```

```
  extensions: [health_check]
  pipelines:
    traces:
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp/dynatrace]
    metrics:
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp/dynatrace]
    logs:
      receivers: [otlp]
      processors: [memory_limiter, resource, batch]
      exporters: [otlphttp/dynatrace]
```

Resource Sizing Guide

Environment	CPU Limit	Memory Limit	Batch Size
Dev/Test	200m	256Mi	500
Staging	500m	512Mi	1000
Production	1000m	1Gi	2000

Running with Environment Variables

```
export DT_ENV_ID="abc12345"
export DT_API_TOKEN="&lt;your-api-token&gt;"

otelcol-contrib --config otel-collector-config.yaml
```

4. Direct SDK Export

Python Direct to Dynatrace

```
import os
from opentelemetry import trace
from opentelemetry.sdk.trace import TracerProvider
from opentelemetry.sdk.trace.export import BatchSpanProcessor
from opentelemetry.exporter.otlp.proto.http.trace_exporter import
OTLPSpanExporter
from opentelemetry.sdk.resources import Resource

# Configuration
DT_ENV_ID = os.environ["DT_ENV_ID"]
DT_API_TOKEN = os.environ["DT_API_TOKEN"]

# Resource with service name
resource = Resource.create({
    "service.name": "my-python-app",
    "service.version": "1.0.0",
    "deployment.environment.name": "production"
})

# Setup tracer
provider = TracerProvider(resource=resource)
exporter = OTLPSpanExporter(
    endpoint=f"https://{DT_ENV_ID}.live.dynatrace.com/api/v2/otlp/v1/traces",
    headers={"Authorization": f"Api-Token {DT_API_TOKEN}"}
)
provider.add_span_processor(BatchSpanProcessor(exporter))
trace.set_tracer_provider(provider)
```

Java with System Properties

```
# Set environment variables first
export DT_ENV_ID="abc12345"
export DT_API_TOKEN="&lt;your-api-token&gt;"

# URL-encode the token for the header (space becomes %20)
java -javaagent:opentelemetry-javaagent.jar \
  -Dotel.service.name=my-java-app \
  -
Dotel.exporter.otlp.endpoint=https://${DT_ENV_ID}.live.dynatrace.com/api/v2/ot
lp \
  -Dotel.exporter.otlp.headers="Authorization=Api-Token%20${DT_API_TOKEN}" \
  -jar app.jar
```

Node.js

```
const { NodeSDK } = require('@opentelemetry/sdk-node');
const { OTLPTraceExporter } = require('@opentelemetry/exporter-trace-otlp-
http');
const { resourceFromAttributes } = require('@opentelemetry/resources');  // JS
SDK 2.x+: the Resource class is no longer exported

const sdk = new NodeSDK({
  resource: resourceFromAttributes({
    'service.name': 'my-node-app',
  }),
  traceExporter: new OTLPTraceExporter({
    url:
`https://${process.env.DT_ENV_ID}.live.dynatrace.com/api/v2/otlp/v1/traces`,
    headers: {
      'Authorization': `Api-Token ${process.env.DT_API_TOKEN}`,
    },
  }),
});

sdk.start();
```

> **Sources:** Upgrade to OpenTelemetry JS SDK 2.x (OpenTelemetry GitHub) — *"The `Resource` class is no longer exported; instead use new utility functions."*

5. Entity Mapping

How Dynatrace Maps OTel Data

OTel Resource Attribute	Dynatrace Entity	Notes
`service.name`	Service	Names the service; without it OTel SDKs report `unknown_service`
`service.namespace`	—	Scopes `service.name`; two services may share a name in different namespaces
`host.name`	Host	Links to host entity
`k8s.namespace.name`	K8s namespace	Links to K8s entities
`k8s.pod.name`	K8s pod	Links to the pod

For the exact attributes Dynatrace uses to map OTLP data to entities, see the enrichment and resource-attribute guidance on the OTLP API (DT docs) page — *"make sure your traces have the correct mapping resource attributes set."*

Essential Resource Attributes

Always set these for proper entity mapping:

```
resource = Resource.create({
    # Required
    "service.name": "checkout-api",

    # Recommended
    "service.version": "1.2.3",
    "service.namespace": "ecommerce",
    "deployment.environment.name": "production",

    # For K8s
    "k8s.namespace.name": "checkout",
    "k8s.pod.name": "checkout-api-abc123",
    "k8s.deployment.name": "checkout-api",
})
```

6. Span Attributes for Dynatrace

Dynatrace fields you should not set yourself

Fields in the `dt.*` namespace are populated by Dynatrace. `dt.span.type` and `dt.source`, which older material lists as attributes to set, have no row in the semantic dictionary, and `dt.entity.*` fields are `deprecated`. Send standard semantic-convention attributes and let Dynatrace derive its own fields.

> **Dictionary:** no row for `dt.span.type` or `dt.source`, read 10/02/2026 (control: `dt.openpipeline.source` returned a row).

Semantic Conventions Dynatrace Uses

Convention	Dynatrace Feature
`http.*`	HTTP service detection
`db.*`	Database call analysis
`messaging.*`	Message queue tracking
`rpc.*`	RPC call tracking

Example with Rich Attributes

```
with tracer.start_as_current_span("checkout") as span:
    # Standard semantic conventions
    span.set_attribute("http.request.method", "POST")
    span.set_attribute("url.full", "https://api.example.com/checkout")
    span.set_attribute("http.response.status_code", 200)
    span.set_attribute("http.route", "/checkout")

    # Business context
    span.set_attribute("order.id", order_id)
    span.set_attribute("order.total", order_total)
```

Complex attribute types (SaaS 1.344)

SaaS 1.344+ (staged tenant rollout from 07/29/2026). Per the release notes, *"Dynatrace now preserves complex attribute values on ingested OTLP spans. Nested key-value list attributes (maps) are now supported, keeping their structure intact."* (SaaS 1.344 release notes (DT docs))

Verify the release has reached your tenant before you start exporting nested attributes. How a pre-1.344 tenant treats a nested attribute is not something to discover from production trace data; send one span from a test service and confirm the structure arrives intact.

Flat scalar attributes are not superseded. They remain the right default for anything you filter, group, or alert on — they are cheaper to query and they keep cardinality legible (see **OTEL-04 § 8 — Attribute Guidelines**). A nested structure is opaque to a `by:{...}` clause in the way a flat key is not.

Attribute shape	Use when	Example
Flat scalar	The value is a query dimension — you filter, group, sort, or alert on it	`order.tier = "gold"`, `payment.provider = "stripe"`
Nested array / map	The structure *is* the payload — you carry it for context and read it on a single span	A validation-error list, a resolved routing table, a request body echo

Reach for nesting only in the second case. If you find yourself planning to aggregate on something inside a nested attribute, promote that value to its own flat scalar attribute alongside the structure.

7. OTLP Metric Dimensions

Advanced OTLP Metric Dimensions

An opt-in setting changes which attributes become metric dimensions. With **Advanced OTLP metric dimensions** on, *"Dynatrace ingests all resource, scope, and data-point attributes as metric dimensions by default, except for attributes on the Deny list: all attributes."* The same page notes that *"dimension keys that were previously normalized*

are now ingested as-is", and that *"Relaxed ingestion limits apply when this feature is enabled."*

It also changes how explicit-bucket histograms are stored: with the setting on they map to a Dynatrace **Histogram**; with it off, to a **Counter**.

Before enabling it, list the dimensions your SLOs, alerts and dashboards filter on (including any `dt.entity.service` filters on OTLP metrics). Afterwards, check that each still exists with the same key and case.

Scope Attributes as Dimensions

With Advanced OTLP metric dimensions on, *"Meter name and version are always added as dimensions (otel.scope.name and otel.scope.version) for Grail-ingested metrics, regardless of the Add Meter name and version as metric dimensions setting."* With it off, the **Add Meter name and version as metric dimensions** setting controls them.

> **Sources:** Configure OTLP metrics ingestion (DT docs), About OTLP metrics ingest (DT docs) — instrument mapping table.

OTLP Auto-Configuration for Kubernetes

Dynatrace Operator (v1.8+) supports **automatic OTLP exporter configuration** for K8s workloads:

Feature	Details
Auto-injection	Environment variables injected into application pods at startup
Routing	Traffic routes through in-cluster ActiveGate if available
Control	Per-pod opt-in/out via `otlp-exporter-configuration.dynatrace.com/inject` annotation
Docs	OTLP Auto-Config

For more details, see Configure OTLP Metrics Ingestion.

```
// Verify OTel traces are arriving over OTLP
// dt.openpipeline.source identifies the ingest path; otel.scope.name also
appears on OneAgent spans.
fetch spans, from:-1h
| filter dt.openpipeline.source == "/api/v2/otlp/v1/traces"
| summarize count = count(), by:{service.name, otel.scope.name}
| sort count desc
| limit 20
```

```
// Check recent OTel spans (OTLP-ingested)
fetch spans, from:-1h
| filter dt.openpipeline.source == "/api/v2/otlp/v1/traces"
| fields start_time, service.name, span.name, duration
| sort start_time desc
| limit 30
```

```
// View OTel data by instrumentation scope (OTLP-ingested spans)
fetch spans, from:-1h
| filter dt.openpipeline.source == "/api/v2/otlp/v1/traces"
| summarize count = count(), by:{otel.scope.name}
| sort count desc
| limit 10
```

8. Verification

Verify Data in Dynatrace

1. **Services**: Go to **Services** - OTel services appear with OTel icon
2. **Traces**: Go to **Distributed Traces** - Filter by service
3. **Metrics**: Go to **Metrics** - Search for custom metric names
4. **Logs**: Go to **Logs & Events** - Filter by trace_id

Troubleshooting Checklist

Check	How
Token scopes	Settings > Access tokens
Collector logs	`otelcol-contrib --config config.yaml`

Check	How
Network connectivity	`curl -v https://{env}.live.dynatrace.com/api/v2/otlp`
service.name set	Check resource attributes

Common Issues

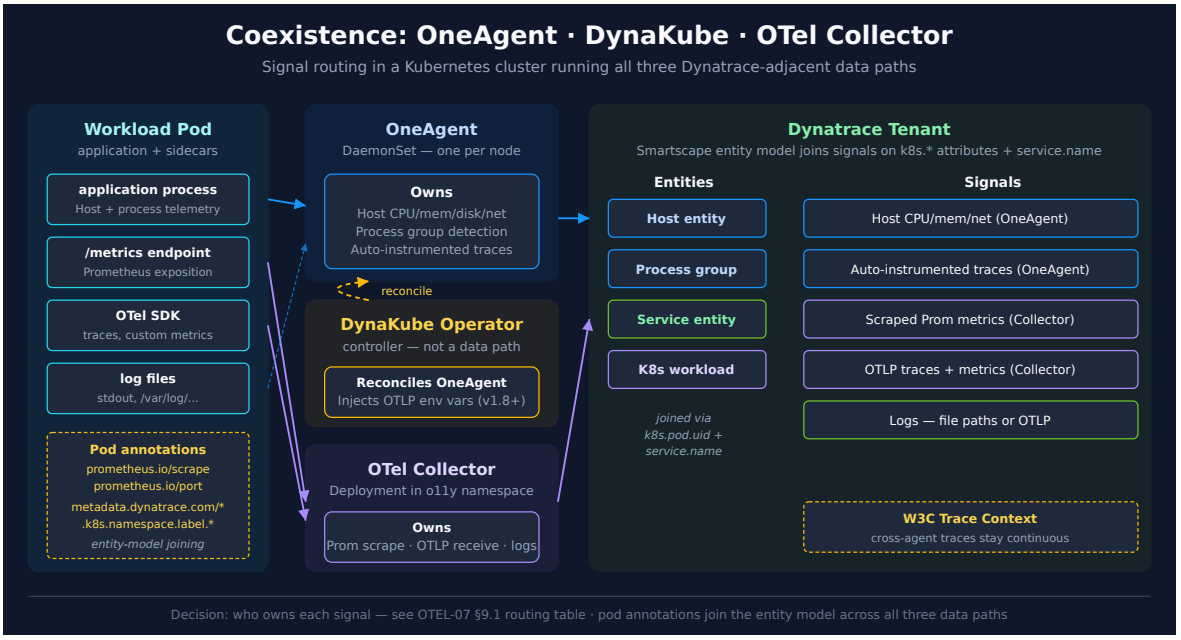
Issue	Cause	Fix
401 Unauthorized	Invalid token	Check token, regenerate
403 Forbidden	Missing scope	Add required scopes
Service shows as `unknown_service`	service.name missing	Add the resource attribute
Request rejected over payload format	JSON payload — *"Binary format must be used"*	Send binary protobuf (`OTEL_EXPORTER_OTLP_PROTOCOL=http/protobuf`)
Partial data	Network timeout	Check batch size, retry config

9. Coexistence with OneAgent and DynaKube

In Kubernetes you may have three Dynatrace-adjacent data paths active at once:

- **OneAgent** — host-process auto-instrumentation, deployed by the DynaKube operator.
- **OTel Collector** — your own deployment, scraping Prometheus targets or accepting OTLP from apps.
- **OTLP auto-config** — DynaKube Operator (v1.8+) can inject OTLP exporter env vars into application pods.

These do not conflict on the wire, but they overlap on what telemetry they produce. The decision is **which agent handles which signal**.



9.1. Signal-Routing Decision Table

In community practice, placement tends to follow the table below — confirm it against your own OneAgent coverage before committing to it:

Signal type	Preferred path	Why
Host CPU / memory / disk / network	OneAgent (DynaKube)	Auto-discovers; tightly integrated with the Smartscape host entity
Process-level metrics, process group detection	OneAgent where installed; otherwise Collector `hostmetrics` + the OpenTelemetry Host Monitoring extension	OneAgent detects process groups automatically; the extension generates host and process entities from Collector data
Application traces (OneAgent-supported language)	OneAgent	Auto-instrumentation, no code change
Application traces (unsupported language or fine-grained custom)	OTel SDK → Collector → OTLP	OTel covers languages OneAgent does not

Signal type	Preferred path	Why
Prometheus-exposed metrics from third-party / business apps	OTel Collector `prometheus` receiver → OTLP	Scales horizontally with a Target Allocator; a Dynatrace Prometheus extension (run on a OneAgent host or remotely on an ActiveGate) is the alternative when you want host-context enrichment
Custom application metrics	OTel SDK → Collector → OTLP	Same as Prometheus path
Logs from OneAgent-instrumented hosts	OneAgent	Auto-detection from log paths
Logs from OTel-only apps	OTel SDK → Collector → OTLP	OneAgent does not pick them up without the agent installed

9.2. Entity Model Joining via `metadata.dynatrace.com/*`

When the OTel Collector scrapes a pod, Dynatrace does not automatically know how to link the resulting metrics to existing entities (the service detected by OneAgent, the K8s deployment in DynaKube's view). The `k8sattributes` processor plus a pod-annotation convention solves this:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: myapp
spec:
  template:
    metadata:
      annotations:
        # Standard Prometheus scrape contract (see OTEL-05 §6.2)
        prometheus.io/scrape: "true"
        prometheus.io/port: "8000"

        # Dynatrace entity-metadata hints — picked up by k8sattributes
processor
        metadata.dynatrace.com/k8s.namespace.label.domain: sandbox
        metadata.dynatrace.com/k8s.namespace.label.otel.service.name: myapp
```

The collector's `k8sattributes` processor pulls these annotations into the resource attributes of every scraped metric. A downstream `transform` processor can then merge the JSON blob into top-level attributes. Result: scraped metrics carry `service.name = "myapp"` matching what OneAgent or the OTel SDK produced for the same workload — and Dynatrace's entity model joins them.

9.3. Counter Temporality — `cumulativetodelta`

Prometheus emits **cumulative** counters that only ever increase. Dynatrace requires **delta** temporality — the per-interval change: *"The Dynatrace backend exclusively works with delta values and requires the respective aggregation temporality."* Its instrument mapping gives a cumulative Counter no Dynatrace metric type, so Prometheus counters must be converted before export. The collector's `cumulativetodelta` processor performs the conversion in-pipeline:

```
processors:
  cumulativetodelta:
    max_staleness: 3m
```

For full pipeline shape including this processor see **OTEL-05 §6.5 Full Pipeline**.

> **Gotcha — processors that create sums.** Prometheus is not the only source of sums Dynatrace will not ingest. A processor that builds new sum metrics sets their temporality itself. Two examples from the full Dynatrace Bindplane OTel Collector build (`dynatrace-bindplane-otel-contrib` v1.2.0): `metricstatsprocessor` emits its statistics as **cumulative** sums when the input is a sum, and `metricextractprocessor` with `metric_type: counter_double` or `counter_int` creates sums with **no temporality set**. Cumulative sums are not ingested — *"Cumulative histograms aren't ingested (similarly to cumulative counters)."* What Dynatrace does with an unspecified temporality is not documented, so do not send it.
>
> - After `metricstatsprocessor` , add `cumulativetodelta` (it ships in the Dynatrace OTel Collector as well). It converts only monotonic sums — *"Non-monotonic sums are excluded."* — so check the output for non-monotonic inputs.
> - For `metricextractprocessor` , keep a gauge type (`gauge_double` , the default, or `gauge_int`). `cumulativetodelta` skips any sum whose temporality is not cumulative, so it cannot fix a `counter_*` metric.

9.4. Auth Scheme Reminder

OTLP from a Collector uses `Authorization: Api-Token ${TOKEN}` (classic) or `Authorization: Bearer ${TOKEN}` (Platform Token). DynaKube's data-ingest token follows the same rule — match the scheme to the token prefix. See **§2 Authentication Setup → Token Type and Auth Scheme** for the matrix.

9.5. Trace Context Propagation

OneAgent and OTel both support W3C Trace Context, so a request that crosses agent boundaries produces a single trace in Dynatrace:

```
Service A (OneAgent) → Service B (OTel) → Service C (OneAgent)
                ↓                  ↓
            Dynatrace traces show complete path
```

Configure OTel to use W3C propagation explicitly — some SDKs default to other formats:

```
from opentelemetry.propagate import set_global_textmap
from opentelemetry.propagators.composite import CompositePropagator
from opentelemetry.propagators.w3c.traceparent import W3CTraceparentPropagator
from opentelemetry.propagators.w3c.tracestate import W3CTracestatePropagator

set_global_textmap(CompositePropagator([
    W3CTraceparentPropagator(),
    W3CTracestatePropagator()
]))
```

9.6. Managing Collector Fleets with Bindplane

Once a team runs more than a handful of OTel Collectors, the work shifts from writing one configuration to keeping many of them consistent. Dynatrace documents **Bindplane** for that job: *"a unified telemetry pipeline to manage OpenTelemetry Collector fleets centrally"*. It manages collectors remotely over OpAMP, the open protocol for collector management, and it fits the open-foundation model this series follows. The same fleet can *"send data to multiple backends simultaneously, including Dynatrace and Splunk."* Dynatrace announced an agreement to acquire Bindplane on 04/08/2026.

Question	Where Bindplane fits
Many Collectors drifting apart in configuration	Central fleet management from one control plane
Telemetry that must reach Dynatrace **and** another backend	Multi-destination routing from the same Collectors
Deployment model	Cloud or self-hosted

Practical notes:

- **Bindplane does not change the coexistence decisions above.** It manages *Collectors*; the signal-routing table in §9.1 and the Span Sensor vs OTLP rule still decide which component emits which signal.
- **Do not write into `dt.bindplane.*`.** From SaaS 1.346 that namespace is *"reserved exclusively for Bindplane"*. A processing rule that sets those fields should move to a namespace you own.
- **Bindplane is optional.** It helps operate many Collectors; it is not a prerequisite for sending OTel data to Dynatrace. As with any SaaS sprint, rollout to tenants is staged — check the tenant version.
- **Know which Collector Bindplane runs, and what its support covers.** *"Bindplane uses the Dynatrace Bindplane OTel Collector by default, or you can bring your own OpAMP-compatible OTel Collector."* The Dynatrace OTel Collector support statement in §1 covers *"only the components and their versions that are included in the Dynatrace OTel Collector."* The full Dynatrace Bindplane OTel Collector ships components the Dynatrace OTel Collector does not — `metricstatsprocessor` and `metricextractprocessor` from `dynatrace-bindplane-otel-contrib`, for example — and those fall outside that statement.

> **Sources:**
>
> - k8sattributes processor (OpenTelemetry Collector Contrib GitHub)
> - Cumulative to Delta processor (OpenTelemetry Collector Contrib GitHub) — *"Non-monotonic sums are excluded."*
> - Configure OTLP Metrics (DT docs)
> - Dynatrace Operator (Dynatrace GitHub)
> - About OTLP metrics ingest (DT docs) — *"The Dynatrace backend exclusively works with delta values and requires the respective aggregation temporality."*; *"Cumulative histograms aren't ingested (similarly to cumulative counters)."*

- Bindplane (DT docs) — *"Bindplane is a unified telemetry pipeline to manage OpenTelemetry Collector fleets centrally"*; *"Collectors can send data to multiple backends simultaneously, including Dynatrace and Splunk."*; *"Bindplane uses the Dynatrace Bindplane OTel Collector by default, or you can bring your own OpAMP-compatible OTel Collector."*
- Dynatrace OTel Collector (DT docs) — *"This support covers only the components and their versions that are included in the Dynatrace OTel Collector."*
- Dynatrace OTel Collector manifest (Dynatrace GitHub)
- Dynatrace Bindplane OTel Collector manifest v1.2.0 (Dynatrace GitHub)
- metricstatsprocessor v1.2.0 (Dynatrace GitHub)
- metricextractprocessor v1.2.0 (Dynatrace GitHub)
- **Code read (10/29/2026):** `metricstatsprocessor/processor.go` L282–284 sets `AggregationTemporalityCumulative` on every sum it emits; `metricextractprocessor/processor_ottl.go` L110–111 creates a sum for `counter_*` types with no temporality call; `cumulativetodeltaprocessor/processor.go` (contrib v0.163.0) skips any sum whose temporality is not cumulative. The Dynatrace Bindplane OTel Collector manifest (v1.2.0) lists both processors; the Dynatrace OTel Collector manifest (v0.59.0) lists neither and does list `cumulativetodeltaprocessor`.
- **Derived:** §9.6 support boundary combines the Dynatrace OTel Collector support statement with the two manifests.
- Dynatrace to acquire Bindplane (Dynatrace press release, 04/08/2026) — *"today announced it has signed a definitive agreement to acquire Bindplane"*
- What's new in Dynatrace SaaS 1.346 (DT docs) — *"The `dt.bindplane.*` namespace is now reserved exclusively for Bindplane."*
- OpenTelemetry Host Monitoring extension (DT docs) — *"This extension automatically generates entities for hosts and their running processes."*
- Prometheus data source (DT docs) — *"For high-volume Prometheus scraping in Kubernetes, and for new deployments, consider the OpenTelemetry Collector"*

Summary

In this notebook, you learned:

- Dynatrace OTLP endpoints (HTTP only — gRPC requires Collector)

- Dynatrace OTel Collector distribution for production deployments, what Dynatrace Support covers, and Contrib for test setups
- API token creation with required scopes; classic `Api-Token` vs Platform `Bearer` auth schemes
- Complete Collector configuration
- Direct SDK export configuration
- Entity mapping and resource attributes
- Span attributes that enhance Dynatrace analysis
- OTLP metric dimensions changes and auto-configuration
- Verification and troubleshooting
- Coexistence patterns across OneAgent, DynaKube, and the OTel Collector — signal routing, entity-model joining via `metadata.dynatrace.com/*`, counter temporality, and processors that create sums Dynatrace will not ingest
- Managing OTel Collector fleets centrally with Bindplane, and the reserved `dt.bindplane.*` namespace

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References

- Dynatrace OpenTelemetry
- OTLP API Endpoints
- Dynatrace Collector
- Configure OTLP Metrics
- OTLP Auto-Config for K8s
- Ensure Success with OTel
- Bindplane

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.